Range

1

a 13　　b 31　　c 2.2981

2 23

3 11

4

a 5　　b 72

5 4

Interquartile range

6

a

i 7　　ii 13　　iii 15　　iv 7

v 16　　vi 9

b

i 7　　ii 1　　iii 18　　iv -3

v 9　　vi 12

c

i 7　　ii 38　　iii 38　　iv 33

v 48　　vi 15

d

i 7　　ii 2.9　　iii 3.6　　iv 1.6

v 3.5　　vi 1.9

e

i 15　　ii 35　　iii 27　　iv 27

v 40　　vi 13

f

i 19　　ii 28　　iii 43　　iv 13

v 38　　vi 25

7

a 15　　b 15　　c 9　　d 16

e 7

8

a

Score	Frequency	Cumulative frequency
20	5	5
30	2	7
40	5	12
50	3	15
60	4	19

b 40 c 30 d 50

e 20

Quantiles, percentiles and deciles

9

a 5.10 b 4.5 c 3 d 8

e 2 f 8 g 4 h 9

10

a 3 b 5 c 2 d 5

11

a 10 b 25 c 10.9 d 20.2

12

a 4 b 84

13

a 75.5 b 42 c 59 d 29

e $\frac{161}{2}$ f 37 g 77

14 $p = 90$

15 $p = 74$

Standard deviation

16
a 9.57 b 5.30

17
a 47.07 b 25.06 c 628.0

18 1.36

19
a 4 b 54.14 c 1.41 d 1.99

20 1.41

21
a

Class	Class centre (x)	f	fx
$1-9$	5	8	40
$10-18$	14	6	84
$19-27$	23	4	92
$28-36$	32	6	192
$37-45$	41	8	328
Total		32	736

b 23.00

c 13.87

d Higher standard deviation, because the grouped data uses the midpoint as an average of the scores which most likely has a lower spread than the actual scores.

22
a 4 b 137 c 137 d 136.36
e 1.3

Applications

23

a One person b 15 people c 16 names d 10 names

e 6

24

a 36 beats per minute b 360 seconds longer

25

a 47 b 10 c 42.55% d 8

e 12 f 36

26

a $15 b $16.50 c International

27 Country A

28

a 0.36

b Increase the mean and increase the standard deviation.

c 0

29

a **B** b i 95% ii 164 cm

30

a 11 b 3

c 2.9 d Standard deviation

31 50%

32

a 3 games b 3.5 c 2.95 d 1.75

e 3.06

33

a The Red class, because their mean was 56.4 which was higher than the Blue class's mean of 56.3.

b The Blue class, because their standard deviation was 3.74 which was lower than the Red class's standard deviation of 4.98.

34

a $3911 million   b Higher   c Unchanged   d Higher

35

a Decreased by 12

b Increased by 4.57

c Decreased by 8

d Increased by 20

36

a

Height of step	Data	Slow	Medium	Fast
Short step	Avg. heart rate	90.0	105.5	126.0
	Standard deviation of heart rate	1.0	0.5	2.0
Tall step	Avg. heart rate	98.0	127.0	129.5
	Standard deviation of heart rate	2.0	2.0	2.5

b A short step at a fast stepping rate.

c A short step at a medium stepping rate.